AI ENGINEER REAL TALK

The AI Engineer Roadmap

Free GitHub repos, YouTube playlists and a step-by-step plan to become an AI Engineer in India.

The truth first. An AI Engineer is a backend developer who knows AI. You will spend less time training models and more time on data, APIs, cost, speed and accuracy. This roadmap is built for that job.

Nikhil Tiwary | AI Engineer | M.Tech AI, IIT Ropar
nikhiltiwary.in

All links checked in November 2026. All resources are free.

START HERE

How to use this roadmap

- Go in order. Each phase uses the one before it.
- Pick **one** resource per topic and finish it. Collecting links is not learning.
- Build the project at the end of every phase and push it to GitHub. Projects get you interviews, not certificates.
- Spend 1 to 2 hours a day. Total time is about 5 to 6 months.
- Stuck for more than 30 minutes? Ask AI to explain, then write the code yourself.

The full plan at a glance

#	Phase	Time	You build
01	**Python and Git**	3 to 4 weeks	A command line expense tracker that saves data to a JSON file.
02	**Data handling**	2 to 3 weeks	Take a messy public CSV (job posts, movies, anything).
03	**Backend with FastAPI**	4 to 5 weeks	A notes API with login, Postgres and Docker.
04	**LLM basics, APIs and prompting**	2 to 3 weeks	A FastAPI endpoint that takes any article and returns a summary as clean JSON (title, 3 points, tags).
05	**RAG (chat with your data)**	3 weeks	A chat-with-your-PDFs app.
06	**Agents and MCP**	3 weeks	An agent that reads your GitHub repos through an MCP server and writes a project summary for your resume.
07	**Production: cost, speed, accuracy**	3 weeks	Take your Phase 5 app.

Big maps if you want the full picture

Resource	Why use it
roadmap.sh: AI Engineer and Backend roadmaps github.com/kamranahmedse/developer-roadmap	Visual roadmaps. Open roadmap.sh/ai-engineer and roadmap.sh/backend.
Shubhamsaboo/awesome-llm-apps github.com/Shubhamsaboo/awesome-llm-apps	Hundreds of working LLM apps. Great for project ideas.

PHASE 01 | 3 TO 4 WEEKS

Python and Git

Everything else is built on this. You do not need to be an expert. You need to be comfortable.

What to learn

- Variables, loops, functions, lists, dicts
- OOP basics: classes and objects
- Files, JSON, error handling (try / except)
- Virtual environments and pip
- Git basics: commit, branch, push, pull request

Resources

Resource	Why use it
Corey Schafer: Python Tutorials (YouTube, English) www.youtube.com/playlist?list=PL-osiE80TeTt2d9bfVyTiXJA-UTHn6WwU	Clear and calm. Start from video 1.
Chai aur Code: Python series (YouTube, Hindi) www.youtube.com/@chaiaurcode	Same basics in Hindi. Code is in the chai-aur-python repo.
CodeWithHarry: Python in Hindi (YouTube) www.youtube.com/playlist?list=PLu0W_9lII9agICnT8t4iYVSZ3eykIAOME	Another Hindi option. Pick one channel, not all three.

Build this: A command line expense tracker that saves data to a JSON file. Push it to GitHub.

PHASE 02 | 2 TO 3 WEEKS

Data handling

AI can help you clean data. But understanding the data is your job. Bad data in, bad answers out.

What to learn

- NumPy and pandas basics
- Cleaning: missing values, duplicates, wrong formats
- Reading CSV, JSON, PDFs and text files
- SQL basics: SELECT, JOIN, GROUP BY

Resources

Resource	Why use it
Python Data Science Handbook (GitHub) github.com/jakevdp/PythonDataScienceHandbook	Free book as notebooks. Read the NumPy and pandas chapters.

Build this: Take a messy public CSV (job posts, movies, anything). Clean it and write 5 insights in a notebook.

PHASE 03 | 4 TO 5 WEEKS

Backend with FastAPI

This is the 40% nobody tells you about. An AI engineer is a backend developer who knows AI.

What to learn

- How REST APIs work: GET, POST, status codes
- FastAPI + Pydantic for request and response models
- PostgreSQL with SQLAlchemy, migrations with Alembic
- Auth (JWT), testing with pytest
- Docker basics, async in Python
- Basic system design: caching, queues, rate limits

Resources

Resource	Why use it
freeCodeCamp: Python API Development (YouTube) www.youtube.com/watch?v=0sOvCWFmrtA	19 hours. FastAPI, Postgres, SQLAlchemy, Alembic, pytest, Docker. Best single course for this phase.
Course code: Sanjeev-Thiyagarajan/fastapi-course github.com/Sanjeev-Thiyagarajan/fastapi-course	Code for the course above. Compare when you get stuck.
zhanymkanov/fastapi-best-practices github.com/zhanymkanov/fastapi-best-practices	How real teams structure FastAPI projects.
fastapi/full-stack-fastapi-template github.com/fastapi/full-stack-fastapi-template	Official template. Read it to see a production setup.
donnemartin/system-design-primer github.com/donnemartin/system-design-primer	Caching, load balancing, queues. Read slowly, one topic a day.
ByteByteGoHq/system-design-101 github.com/ByteByteGoHq/system-design-101	Same ideas as simple diagrams.

Build this: A notes API with login, Postgres and Docker. Write tests. Deploy it free (Render, Railway or similar).

PHASE 04 | 2 TO 3 WEEKS

LLM basics, APIs and prompting

You will call models through APIs every day. Learn how they work and how to talk to them.

What to learn

- What tokens, context window and temperature mean
- Calling OpenAI, Claude and Gemini APIs from Python
- Prompt engineering: clear instructions, examples, roles
- Structured output (JSON) and function / tool calling
- Streaming responses

Resources

Resource	Why use it
microsoft/generative-ai-for-beginners github.com/microsoft/generative-ai-for-beginners	Free course with lessons and code. Good first stop.
anthropics/prompt-eng-interactive-tutorial github.com/anthropics/prompt-eng-interactive-tutorial	Hands-on prompt engineering in notebooks.
dair-ai/Prompt-Engineering-Guide github.com/dair-ai/Prompt-Engineering-Guide	Reference for prompting techniques.
openai/openai-cookbook github.com/openai/openai-cookbook	Ready code examples for real tasks.
anthropics/claude-cookbooks github.com/anthropics/claude-cookbooks	Same for Claude: tool use, JSON, RAG.

Build this: A FastAPI endpoint that takes any article and returns a summary as clean JSON (title, 3 points, tags).

PHASE 05 | 3 WEEKS

RAG (chat with your data)

Most company AI products are RAG. The model answers using your documents, not only its memory.

What to learn

- Embeddings and vector databases (Pinecone, Qdrant, pgvector)
- Chunking documents the right way
- Retrieval, re-ranking, hybrid search
- Checking if answers are grounded in the documents

Resources

Resource	Why use it
NirDiamant/RAG_Techniques github.com/NirDiamant/RAG_Techniques	Every RAG technique with a notebook. Start simple, go deeper later.
DataTalksClub/llm-zoomcamp github.com/DataTalksClub/llm-zoomcamp	Free course. Build a full RAG app step by step.

Build this: A chat-with-your-PDFs app. Upload PDFs, ask questions, show which page the answer came from.

PHASE 06 | 3 WEEKS

Agents and MCP

Agents let the model use tools: search, databases, APIs. MCP is the standard way to plug tools in.

What to learn

- Tool calling and the agent loop
- LangGraph: state, nodes, memory, human-in-the-loop
- MCP: build a simple MCP server and connect it
- When NOT to use an agent (simple code is often better)

Resources

Resource	Why use it
LangChain Academy: Introduction to LangGraph academy.langchain.com	Free official course. Notebooks: github.com/langchain-ai/langchain-academy
microsoft/ai-agents-for-beginners github.com/microsoft/ai-agents-for-beginners	Free lessons on agent patterns.
huggingface/agents-course github.com/huggingface/agents-course	Free course with exercises.
microsoft/mcp-for-beginners github.com/microsoft/mcp-for-beginners	MCP from zero.
modelcontextprotocol/servers github.com/modelcontextprotocol/servers	Real MCP servers to read and copy patterns from.
NirDiamant/GenAI_Agents github.com/NirDiamant/GenAI_Agents	Many agent examples with code.

Build this: An agent that reads your GitHub repos through an MCP server and writes a project summary for your resume.

PHASE 07 | 3 WEEKS

Production: cost, speed, accuracy

This is what you do at the job. A demo works once. A product works 10,000 times a day.

What to learn

- Latency: streaming, smaller models, parallel calls
- Cost: model routing (cheap model for easy tasks), token limits
- Caching repeated questions
- Retries, timeouts and a backup model when the API fails
- Handling bad JSON from the model
- Evals: test sets, LLM-as-judge, error analysis
- Logging and tracing every call

Resources

Resource	Why use it
chiphuyen/aie-book github.com/chiphuyen/aie-book	Resources from Chip Huyen's AI Engineering book. The best map of this field.
Hamel Husain: Evals FAQ hamel.dev/blog/posts/evals-faq/	Practical answers on how to evaluate LLM apps.
Lenny's Podcast: AI evals with Hamel and Shreya (YouTube) www.youtube.com/watch?v=BsWxPI9UM4c	Live demo of building evals on a real product.
promptfoo/promptfoo github.com/promptfoo/promptfoo	Open-source tool to test prompts and catch regressions.

Build this: Take your Phase 5 app. Add caching, retries, a backup model, cost logging and a 30-question eval set. Write the before vs after numbers in the README.

KEEP THIS

The production checklist

Run every AI project through these questions. This is what interviewers and real teams ask.

Problem	What you do
A call takes 8 seconds?	Stream the answer, use a smaller model, run calls in parallel.
API bill keeps growing?	Cheap model for easy tasks, big model only for hard ones. Cut extra tokens.
Same question comes again and again?	Cache it. Do not pay twice.
Model wrote a story instead of JSON?	Use structured output, validate with Pydantic, retry on failure.
API is down or rate limited?	Retries with backoff, timeouts and a backup model.
Model is confidently wrong?	Build an eval set, measure accuracy, read the failures yourself.
Something broke in production?	Log every prompt, response, latency and cost so you can trace it.

OPTIONAL

Want to train models? That is the ML Engineer path

Model training and research is a different role. In India these jobs are few and many need an MS or PhD. If this excites you, do it after Phase 4, alongside the rest.

Resource	Why use it
Andrej Karpathy: Neural Networks Zero to Hero (YouTube) www.youtube.com/playlist?list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ	Build neural networks and GPT from scratch. About 19 hours.
mlabonne/llm-course github.com/mlabonne/llm-course	LLM fundamentals, fine-tuning and the full LLM engineer path.

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